Exact Coset Sampling for Quantum Lattice Algorithms

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Abstract

We revisit the post-processing phase of Chen's Karst-wave quantum lattice algorithm [Chen, 2024] in the Learning with Errors (LWE) parameter regime. Conditioned on a transcript E, the post-Step 7 coordinate state on $(\mathbb{Z}_M)^n$ is supported on the shifted grid line

$$\{\, j\Delta + \boldsymbol{v}^*(E) + M_2\boldsymbol{k} \bmod M : j \in \mathbb{Z},\ \boldsymbol{k} \in \mathcal{K} \,\},$$

where $\Delta = 2D^2\mathbf{b}$ and $M = 2M_2 = 2D^2Q$ with Q odd. The amplitudes contain a quadratic chirp $\omega_Q^{-j^2}$ and an unknown run-dependent offset $\boldsymbol{v}^*(E)$. We show that Chen's Steps 8-9 can be replaced by a single routine, Step $9^\dagger$: measure $\tau := X_1 \bmod D^2$, apply a τ-dependent quadratic phase on the first coordinate register to remove the chirp, and then apply $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the coordinate registers. The chirp removal uses a canonical gauge promise: conditioned on E, the offset $v_1^*(E) \bmod M_2$ equals the centered lift $\mathrm{ctr}(\tau)$. Under explicit front-end conditions (AC1-AC5), a Fourier sample $\boldsymbol{u} \in \mathbb{Z}_M^n$ satisfies the resonance $\langle \mathbf{b}, \boldsymbol{u}\rangle \equiv 0 \pmod{Q}$ with probability $1 - \mathrm{negl}(n)$, and conditioned on resonance the reduced sample $\boldsymbol{u} \bmod Q$ is exactly uniform on the dual hyperplane. This yields exact recovery of $\mathbf{b} \bmod Q$ and hence the embedded LWE secret.

1 Introduction

Fourier sampling quantum algorithms for lattice problems prepare a structured superposition. A Fourier transform of this state then reveals short modular linear relations. Chen's windowed quantum Fourier transform (QFT) with complex Gaussian windows [Chen, 2024] fits this pattern. The n coordinate registers of the post-Step 7 state $|\widetilde{\varphi}_7\rangle$ live in $\mathbb{Z}_M$, with $M = 2M_2$. The one-shot routine in Algorithm 2 performs the run-local phase correction described below and then applies $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ directly to these registers. The post-Step 7 state has affine support

$$|\widetilde{\varphi}_7\rangle = \sum_{j\in\mathbb{Z}} \sum_{\boldsymbol{k}\in\mathcal{K}} \alpha_E(j,\boldsymbol{k})\, \omega_Q^{-j^2}\, i^{\|\boldsymbol{k}\|^2}\, |\mathbf{X}(j) + M_2\boldsymbol{k}\rangle_M\,, \tag{1.1}$$

for an outcome-dependent amplitude profile $\alpha_E(j,\boldsymbol{k})$ and an index set $\mathcal{K} \subseteq \{0\} \times \mathbb{Z}^{n-1}$ that comes from the grid-state construction. When it is clear from context, we suppress the transcript index E. The phase factor $i^{\|\boldsymbol{k}\|^2}$ comes from Chen's grid-state preparation procedure. We view the coordinate values as elements of $(\mathbb{Z}_M)^n$ throughout. Accordingly, $\mathbf{X}(j) := 2D^2 j\,\mathbf{b} + \boldsymbol{v}^*$ is interpreted componentwise modulo M (as in Chen's post-Step 7 expression), and $\boldsymbol{k}$ records the M_2-shift term

$M_2\boldsymbol{k}$. Reducing the coordinate registers modulo M_2 eliminates this shift and yields the affine congruence $\mathbf{X}(j) \equiv 2D^2 j\,\mathbf{b} + \boldsymbol{v}^* \pmod{M_2}$ used in Additional Condition AC1.

The additional factor $\omega_Q^{-j^2} = \exp(-2\pi i j^2/Q)$ is the Karst-wave chirp. Step $9^\dagger$ removes it using only the first coordinate register. We denote the chirp-free state obtained after this run-local phase correction by $|\varphi_7\rangle$; it satisfies

$$|\varphi_7\rangle = \sum_{j\in\mathbb{Z}} \sum_{\boldsymbol{k}\in\mathcal{K}} \alpha_E(j,\boldsymbol{k})\, i^{\|\boldsymbol{k}\|^2}\ |\mathbf{X}(j) + M_2\boldsymbol{k}\rangle_M\,. \tag{1.2}$$

Additional Condition AC1 records the affine relation $\mathbf{X}(j) = 2D^2 j\,\mathbf{b} + \boldsymbol{v}^*$ for the post-Step 7 state. Additional Condition AC4 fixes a canonical *gauge* for the first-coordinate offset: the preparation circuit is compiled (by an E-controlled basis translation on X_1 by a multiple of D^2) so that measuring $\tau := X_1 \bmod D^2$ determines $v_1^*(E) \bmod Q$ for that run. Under Additional Condition AC1, the first coordinate satisfies the precise congruence

$$X_1(j) \equiv -2D^2 j + v_1^*(E) \pmod{M_2}$$

for every j in the effective window. Write $\Delta := 2D^2\mathbf{b}$ and recall $M = 2M_2 = 2D^2Q$. Since $D^2, Q \mid M$, reducing any physical basis value $x \in \mathbb{Z}_M$ modulo D^2 or Q is unambiguous. In particular,

$$X_1 \bmod D^2 \equiv v_1^* \bmod D^2, \qquad (X_1 \bmod Q) - v_{1,Q} \equiv -2D^2 j \pmod{Q},$$

where under the canonical gauge (AC4) the post-processing computes $v_{1,Q} = v_1^*(E) \bmod Q$ from $\tau = X_1 \bmod D^2$.

In Chen's post-processing one ultimately outputs non-zero Fourier samples in $\mathbb{Z}_{M_2}^n$ (with $M_2 = M/2$) whose reduction modulo $Q = M/(2D^2)$ satisfies a homogeneous dual relation. In our direct-QFT variant we instead measure $\boldsymbol{u} \in \mathbb{Z}_M^n$ and store only the reduction $\boldsymbol{u}_Q := \boldsymbol{u} \bmod Q$. In either view, the goal is to obtain non-zero vectors satisfying

$$\langle \mathbf{b}, \boldsymbol{u}\rangle \equiv 0 \pmod{Q}, \tag{1.3}$$

where $Q := \frac{M}{2D^2}$ is an odd integer. (In Chen [2024] the parameters are additionally arranged so that Q has a convenient factorization, but Step $9^\dagger$ does not use it.) The short vector $\mathbf{b} = (b_1, \ldots, b_n) \in \mathbb{Z}^n$ stores the hidden data and satisfies the normalization $b_1 = -1$. The offset $\boldsymbol{v}^*$ is unknown and can change from run to run. The algorithm then collects $O(n)$ such samples and solves the resulting linear system modulo Q to recover $\mathbf{b} \bmod Q$ and hence the hidden data vector.

Chen's published Steps 8-9 enforce relations of the form Equation (1.3) via additional offset-handling and a domain-extension mechanism that is tailored to a particular parameterization. Step $9^\dagger$ provides an alternative post-processing designed for an access model in which the affine offset $\boldsymbol{v}^*(E)$ may vary across runs and no special factorization of Q is assumed. It uses only the affine structure $\mathbf{X}(j) = j\Delta + \boldsymbol{v}^*$ with $\Delta = 2D^2\mathbf{b}$, the normalization $b_1 = -1$, a run-local chirp cancellation acting on X_1 (Additional Condition AC4), and a direct $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ on the coordinate registers. Under Additional Condition AC5, the resulting outcomes satisfy $\langle \mathbf{b}, \boldsymbol{u}\rangle \equiv 0 \pmod{Q}$ except with negligible failure probability.

When we instantiate the front end with the q-ary lattice $L_q^\perp(\mathbf{A})$ from Chen [2024], the new Step $9^\dagger$ recovers the planted shortest vector $\mathbf{b} = [-1, 2p_1 s^\top, 2p_1 e^\top]^\top$ and hence the LWE secret s and error e in the chosen-secret regime considered there. Combined with the classical reductions of Chen [2024], this yields a quantum algorithm for standard LWE in the same parameter regime, conditional on Additional Conditions AC1-AC5.

2 Background and Access Model

Notations. Let $q \in \mathbb{N}$ and write $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ with representatives in $(-\frac{q}{2}, \frac{q}{2}]$. Vectors are written in bold. We use $\langle \cdot, \cdot \rangle$ for the inner product. For $T \in \mathbb{N}$ we write $\omega_T := \exp(2\pi i/T)$. The *physical* coordinate registers live in $\mathbb{Z}_M^n$, where

$$M := 2M_2 = 2D^2Q, \qquad M_2 := D^2Q.$$

We will repeatedly reduce physical basis values $x \in \mathbb{Z}_M$ modulo M_2, D^2, or Q. Since $D^2, Q \mid M_2 \mid M$, the canonical reduction maps

$$\mathbb{Z}_M \to \mathbb{Z}_{M_2}, \qquad \mathbb{Z}_M \to \mathbb{Z}_{D^2}, \qquad \mathbb{Z}_M \to \mathbb{Z}_Q$$

are well-defined ring homomorphisms.

Modular inner products. For $\boldsymbol{a} \in \mathbb{Z}^n$ and $\boldsymbol{u} \in (\mathbb{Z}_M)^n$ we write

$$\langle \boldsymbol{a}, \boldsymbol{u} \rangle \bmod Q \;:=\; \sum_{i=1}^{n} (a_i \bmod Q) \cdot (u_i \bmod Q) \;\in\; \mathbb{Z}_Q.$$

When we write $\langle \boldsymbol{a}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}$ this is the intended meaning (i.e., the dot product after coordinate-wise reduction $\mathbb{Z}_M \to \mathbb{Z}_Q$).

Representative conventions. We use the standard representatives in $\{0, \ldots, M-1\}$ for $\mathbb{Z}_M$. All coordinate-register kets $|\cdot\rangle_M$ are labelled by elements of $\mathbb{Z}_M$ in these representatives. We also routinely identify $t \in \mathbb{Z}_T$ with its representative in $\{0, \ldots, T-1\}$ when evaluating phases ω_T^t. When we refer to reductions modulo M_2, D^2, or Q, we always mean the canonical ring homomorphisms from $\mathbb{Z}_M$.

We write

$$0\,|\,\mathbb{Z}^{n-1} := \{0\} \times \mathbb{Z}^{n-1}.$$

The physical coordinate block in Chen's Step 7 lives in $\mathbb{Z}_M$ with $M = 2M_2$. When this helps to avoid confusion, we write the reduction modulo M with a subscript $_M$. For a coordinate slice we write $\boldsymbol{x}_{[2..n]} := (x_2, \ldots, x_n)$. Set $\Delta := 2D^2\,\mathbf{b} \in \mathbb{Z}^n$.

We use $\mathrm{poly}(\cdot)$ for an unspecified fixed polynomial and $\mathrm{negl}(n)$ for a negligible function of n.

Idealized unitaries vs. circuit precision. For clarity we treat $\mathrm{QFT}_{\mathbb{Z}_M}$ and the controlled root-of-unity phases used in the chirp correction (AC4) as exact unitaries. In a standard uniform gate set these operations can be implemented to precision $2^{-\mathrm{poly}(n)}$, which perturbs all output distributions by at most $\mathrm{negl}(n)$ in total variation distance. We suppress this approximation issue throughout, since it does not affect any of the modular-algebraic arguments.

Finally, we write $\mathrm{ctr} : \mathbb{Z}_{D^2} \to \mathbb{Z}$ for the centered lift: for $\tau \in \mathbb{Z}_{D^2}$, $\mathrm{ctr}(\tau)$ is the unique integer in $(-D^2/2, D^2/2]$ that is congruent to τ modulo D^2. We use $\mathrm{ctr}(\tau)$ as a canonical integer representative when converting $\tau = X_1 \bmod D^2$ into arithmetic in $\mathbb{Z}_Q$.

At the frontier right before Step 8, and for fixed outcomes E, Lemma 2.8 gives $\mathbf{X}(j) \equiv j\Delta + \boldsymbol{v}^* \pmod{M_2}$ for every j in the effective window.

Parameter identification from Chen [2024]. In Chen's notation, $M = 2(t^2+u^2)$ and $x = D\mathbf{b}$, where $\mathbf{b}$ is the short vector from the LWE-to-lattice reduction (Eq. (12) in Chen [2024]). We follow this identification. Chen's Step 9 defines a relabelled vector $\mathbf{b}^*$ (Eq. (39) in that paper). Our post-processing never uses $\mathbf{b}^*$. We keep all formulas in terms of $\mathbf{b}$ instead.

Under Chen's parameter constraints the quantity $M/(2D^2)$ is an odd integer. We write

$$Q := \frac{M}{2D^2}, \qquad M_2 := \frac{M}{2} = D^2 Q.$$

In Chen's concrete instantiation Q happens to admit a factorisation into pairwise coprime odd factors; the new Step $9^\dagger$ does not use this extra structure and only relies on Q being odd.

Chen uses the letter P for a large loop modulus with $P \approx M^2/2$. We keep this convention and reserve P only for that modulus.

Index domain vs. first coordinate. The preparation uses a loop index j. This index is the discrete time variable of the Karst wave and appears in Chen [2024] as a summation index. It runs over the large modulus P. We denote the register by $J \in \mathbb{Z}_P$.

The first coordinate satisfies

$$X_1 \equiv 2D^2 j\, b_1 + v_1^* \pmod{M_2}.$$

In Chen's setup, v_1^* depends on the measurement outcomes $(y', z', h^\star)$. The experiment cannot be reset in a controlled way to obtain the same outcomes and the same v^*. So we cannot query the state preparation on chosen basis inputs j and then compute $\mathbf{X}(j)$ either classically or into an extra register. We must work with the single quantum state $|\varphi_7\rangle$ that a successful run outputs.

Parameter re-tuning. As in Chen [2024] we run a classical outer loop over polynomially many candidates for $\|\mathbf{b}\|^2$ and, for each guess, set

$$Q = (c{+}1)\|\mathbf{b}\|^2, \qquad M = 2D^2 Q, \qquad M_2 = D^2 Q.$$

The scaling parameter D can then be chosen (odd, $\gcd(D, Q) = 1$, and poly(n)-bounded) to satisfy the geometric separation requirement of the Karst-wave front end (Condition C.6 of Chen [2024]). The only additional analytic requirement used by Step $9^\dagger$ is the one-dimensional spectral concentration property recorded in Additional Condition AC5, which is achieved in Chen's complex-Gaussian regime once the loop-envelope width satisfies $\sigma_J \gtrsim Q \log n$ (Remark 2.5). All subsequent correctness statements are conditional on the existence of a (re-tuned) front-end configuration satisfying Definition 2.1 and Additional Conditions AC1-AC5.

Definition 2.1 (Re-tuned Karst-wave configuration)**.** Take LWE parameters (ℓ, m, q, β) in the regime of Chen [2024] and set $n := 1 + \ell + m$. Let $L_q^\perp(\mathbf{A})$, the planted shortest vector $\mathbf{b}$, and the integers $(p_\eta)_\eta$ be as in Section 3.2 of Chen [2024]. A *re-tuned Karst-wave configuration* is any choice of front-end parameters

$$(c, D, r, s, t, u, P, M, Q)$$

that satisfies the following properties.

(R1) The integers $p_1, \ldots, p_\kappa$ are odd and pairwise coprime. They control the chosen-secret embedding and the parsing of $\mathbf{b}$ in the LWE extraction step, but they do not impose any algebraic constraint on Q. Fix an integer $c = c(n) \in 4\mathbb{Z}$ (typically $c = \Theta(\log^6 n)$ in Chen-style instantiations) and set

$$Q := (c{+}1)\|\mathbf{b}\|^2.$$

In the chosen-secret LWE embedding of [Chen, 2024], $\mathbf{b} = [-1, 2p_1 s^\top, 2p_1 e^\top]^\top$ has one odd coordinate and all remaining coordinates even, hence $\|\mathbf{b}\|^2 \equiv 1 \pmod 4$. Since $c \in 4\mathbb{Z}$, this makes $Q = (c{+}1)\|\mathbf{b}\|^2$ automatically odd. We also require that Q satisfies

$$\|\mathbf{b}\|_\infty \le 2p_1\beta \log n < Q/2,$$

so that $\mathbf{b} \bmod Q$ uniquely determines $\mathbf{b}$ by centered lifting.

(R2) Choose an odd scaling parameter D that is coprime to Q and polynomially bounded in n (for concreteness, one may take the smallest odd integer at least $32\log^2 n$ with $\gcd(D, Q) = 1$). Define

$$u^2 := \|D\mathbf{b}\|^2 = D^2\|\mathbf{b}\|^2, \qquad t^2 := c\,u^2,$$

and set the Karst-wave moduli

$$M := 2(t^2 + u^2) = 2D^2Q, \qquad M_2 := M/2 = D^2Q, \qquad P := M(t^2 + u^2) = \frac{M^2}{2}.$$

Then $2D^2$ is a unit modulo Q.

(R3) Choose $r, s > 0$ so that Chen's Karst-wave constraints C.5-C.7 hold and the resulting geometric width parameter

$$\sigma_{\text{geom}} := \frac{rs^2t}{u\sqrt{r^4 + s^4}}$$

obeys $2\log n < \sigma_{\text{geom}} < D/(4\log n)$. Chen's analysis shows that such choices exist with all parameters bounded by $\text{poly}(n)$; we fix one such choice.

(R4) For the resulting complex-Gaussian front end, let σ_J denote the standard deviation of the one-dimensional envelope $\alpha_E(j)$ along the loop index j (after marginalizing over the grid index and the workspace and conditioning on a transcript E). We restrict to parameter choices satisfying

$$\sigma_J \ge C_{\text{spec}}\, Q \log n$$

for some fixed absolute constant $C_{\text{spec}} > 0$. A lower bound of this form keeps the spectral leakage in Additional Condition AC5 bounded by $\exp(-\Omega(\log^2 n))$.

When we refer to Chen's front end with re-tuned parameters below, we mean an outcome-conditioned circuit family compiled with a fixed choice of parameters that satisfies (R1)-(R4).

2.1 Front-end properties and additional conditions

We now record the structural properties and spectral conditions that the re-tuned front end must satisfy. These conditions allow the LWE solver in Step $9^\dagger$ to work in the exact sampling regime.

AC1. After modulus splitting and center extraction (Steps 5-7 of Chen [2024]) and just before the Step 8 gadget, conditioned on any fixed transcript E the post-Step 7 coordinate block is supported on

$$\{\, j\Delta + \boldsymbol{v}^*(E) + M_2\boldsymbol{k} \bmod M : j \in \mathbb{Z},\ \boldsymbol{k} \in \mathcal{K} \,\},$$

for some index set $\mathcal{K} \subseteq \{0\} \times \mathbb{Z}^{n-1}$, where $\Delta = 2D^2\mathbf{b}$ and the offset $\boldsymbol{v}^*(E) \in (\mathbb{Z}_M)^n$ depends on E but not on j. Reducing modulo M_2 eliminates the $M_2\boldsymbol{k}$ term and yields the affine congruence

$$\mathbf{X}(j) \equiv 2D^2 j\,\mathbf{b} + \boldsymbol{v}^*(E) \pmod{M_2}.$$

This matches the offset-coherence invariant of Chen's deferred-measurement front end (Lemma 2.8). In particular, since $k_1 = 0$ for all $\boldsymbol{k} \in \mathcal{K}$ and $b_1 = -1$, the first coordinate satisfies

$$X_1 \equiv -2D^2 j + v_1^* \pmod{M_2}, \qquad X_1 \bmod D^2 \equiv v_1^* \bmod D^2.$$

AC2. We assume throughout that Q is an odd integer. The post-processing in Step 9† works for an arbitrary composite modulus Q; whenever we invert a residue class we only do so modulo a modulus that is guaranteed to be coprime to that residue (in particular, we only invert $-2D^2$ modulo Q, which is valid by AC2-AC3). In particular, we do not require Q to factor in any prescribed way, and the integers $(p_\eta)_\eta$ from the LWE-to-lattice reduction play no role in the algebraic structure of Q.

AC3. We keep the scaling regime of Chen [2024]. The parameter D is odd, coprime to Q, and we define

$$M := 2D^2Q, \qquad M_2 := M/2 = D^2Q.$$

Since Q is odd (AC2) and $\gcd(D, Q) = 1$, we have $\gcd(2D^2, Q) = 1$, so $2D^2$ is a unit modulo Q. All divisions by D^2 or $2D^2$ that appear in the post-processing should be read as multiplications by the corresponding inverses in $\mathbb{Z}_Q$; we never perform integer division by D^2 or $2D^2$ in $\mathbb{Z}$. For the first coordinate, this gives

$$X_1 \equiv 2D^2 j b_1 + v_1^* \equiv -2D^2 j + v_1^* \pmod{M_2},$$

so that reduction modulo D^2 kills the j-dependent term:

$$X_1 \bmod D^2 \equiv v_1^* \bmod D^2.$$

AC4. The post-Step 7 amplitudes contain the quadratic chirp $\omega_Q^{-j^2}$ in the loop index j. We cancel it with a diagonal unitary acting only on the first coordinate register X_1.

Fix a transcript E. By AC1 and $b_1 = -1$, for every j in the effective window,

$$X_1 \equiv -2D^2 j + v_1^*(E) \pmod{M_2}.$$

Because $k_1 = 0$ for all $\boldsymbol{k} \in \mathcal{K}$, reducing modulo D^2 removes the j-term on every branch:

$$\tau := X_1 \bmod D^2 \equiv v_1^*(E) \bmod D^2.$$

Thus τ is deterministic conditioned on E; measuring it does not disturb the coherent superposition over j and $\boldsymbol{k}$. Let $\bar{v}_1 := \mathrm{ctr}(\tau) \in (-D^2/2, D^2/2]$ and set $v_{1,Q} := \bar{v}_1 \bmod Q \in \mathbb{Z}_Q$.

Under the canonical-gauge requirement below, this $v_{1,Q}$ equals $v_1^*(E) \bmod Q$ for that run.

Canonical gauge (required). Measuring τ determines $v_1^*(E) \bmod D^2$ but leaves an ambiguity by multiples of D^2 inside $\mathbb{Z}_{M_2}$. Equivalently, there is a unique $\lambda(E) \in \mathbb{Z}_Q$ such that

$$v_1^*(E) \equiv \bar{v}_1 + \lambda(E) D^2 \pmod{M_2}.$$

Any correction depending only on τ cancels $\omega_Q^{-j^2}$ only up to a residual run-dependent *linear* phase $\omega_Q^{-\lambda(E)j}$ (up to a global constant), which shifts the resonance class $\langle \mathbf{b}, \boldsymbol{u} \rangle \bmod Q$ and prevents aggregating samples across invocations. We therefore assume a *canonical gauge* in which $\lambda(E) = 0$ for every transcript E, i.e.,

$$v_1^*(E) \equiv \bar{v}_1 \pmod{M_2}.$$

When the preparation circuit is known, this can be enforced by an E-controlled basis translation on X_1 by a multiple of D^2, provided E allows reversible computation of $\lambda(E)$; see Proposition 2.3.

Define the classical inverse $\mathrm{inv} := (-2D^2)^{-1} \in \mathbb{Z}_Q$, which exists by AC2-AC3. For each basis value $x \in \mathbb{Z}_M$ of X_1, write $x_Q := x \bmod Q$ and define

$$j_x := \mathrm{inv} \cdot \big((x_Q - v_{1,Q}) \bmod Q \big) \in \mathbb{Z}_Q.$$

For a branch with loop label j we have $(x_Q - v_{1,Q}) \equiv -2D^2 j \pmod{Q}$, hence $j_x \equiv j \pmod{Q}$. We apply the diagonal unitary $U_{\mathrm{corr}}^{(\tau)}$ on X_1:

$$|x\rangle_{X_1} \longmapsto \omega_Q^{j_x^2} |x\rangle_{X_1},$$

implemented by reversible arithmetic that computes j_x, applies the phase, and uncomputes. Since $j_x^2 \equiv j^2 \pmod{Q}$, this cancels the chirp $\omega_Q^{-j^2}$ on every branch and yields the chirp-free state $|\varphi_7\rangle$ in (1.2).

AC5. For the rephased chirp-free state of AC4 and a fixed transcript E, define for $\boldsymbol{u} \in \mathbb{Z}_M^n$

$$S_E(\boldsymbol{u}) := \sum_{j \in \mathbb{Z}} \sum_{\boldsymbol{k} \in \mathcal{K}} \alpha_E(j, \boldsymbol{k})\, i^{\|\boldsymbol{k}\|^2}\, \omega_M^{j\langle \boldsymbol{u}, \Delta \rangle + M_2 \langle \boldsymbol{u}, \boldsymbol{k} \rangle}.$$

Up to the $\boldsymbol{u}$-dependent global phase $\omega_M^{\langle \boldsymbol{u}, \boldsymbol{v}^* \rangle}$, $S_E(\boldsymbol{u})$ is the (unnormalized) Fourier amplitude of $|\boldsymbol{u}\rangle$ obtained after applying $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the coordinate registers of $|\varphi_7\rangle$.

We view the loop index $j \in \mathbb{Z}_P$ as embedded in $\mathbb{Z}$. We extend $\alpha_E(j, \boldsymbol{k})$ by zero outside an interval $\{|j| \le J_{\max}\}$. The effective window satisfies $J_{\max} = \mathrm{poly}(Q \log n) \ll P$ in the Karst-wave regime below; in particular we assume $J_{\max} < P/2$ so that the embedding of $j \in \mathbb{Z}_P$ into integers is unambiguous on the support of the state. The infinite sum above then matches, up to negligible error, the finite sum that appears in Chen's state.

As in Chen's Karst-wave front end, we assume that conditioned on E the amplitudes factor across the loop index and the grid label:

$$\alpha_E(j, \boldsymbol{k}) = \alpha_E(j)\, \beta_E(\boldsymbol{k}).$$

Define the one-dimensional Fourier transform of the loop envelope

$$\widehat{\alpha}_E(\theta) := \sum_{j \in \mathbb{Z}} \alpha_E(j)\, e^{2\pi i j \theta}.$$

A transcript E is called *good* if the off-resonant Fourier mass on the Q-point grid $\{t/Q : t \in \mathbb{Z}_Q\}$ is negligible:

$$\sum_{t \in \mathbb{Z}_Q \setminus \{0\}} \left|\widehat{\alpha}_E(t/Q)\right|^2 \le \mathrm{negl}(n) \cdot \sum_{t \in \mathbb{Z}_Q} \left|\widehat{\alpha}_E(t/Q)\right|^2. \tag{2.1}$$

We assume that a transcript drawn from the front end is good except with probability $\mathrm{negl}(n)$ over the front-end randomness.

Reliability of the additional conditions. Additional Conditions AC2-AC3 are enforced directly by Definition 2.1. Additional Condition AC1 is the standard offset-coherence invariant of Chen's deferred-measurement front end (Lemma 2.8). Additional Condition AC4 is the only genuinely new access-model promise used by Step 9†: it replaces Chen's Step 8 offset-extraction gadget by a gauge-fixing guarantee (or coherent access to the gauge parameter) sufficient to cancel the quadratic chirp using only X_1. It is enforceable by a compile-time basis translation on the first coordinate whenever the preparation circuit supports reversible computation of the gauge parameter $\lambda(E)$ (Proposition 2.3). Finally, Additional Condition AC5 follows from the complex discrete-Gaussian Karst-wave envelope once $\sigma_J \ge C_{\mathrm{spec}} Q \log n$ (Proposition 2.6).

Lemma 2.2 (Uniformity modulo Q on the dual hyperplane)**.** Assume AC1-AC4 and fix any transcript E. Let μ_E be the distribution of $\boldsymbol{u} \in \mathbb{Z}_M^n$ obtained by applying $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the coordinate registers of $|\varphi_7\rangle$ (after the correction in AC4) and measuring. Define the resonance event and dual hyperplane

$$R := \{\boldsymbol{u} \in \mathbb{Z}_M^n : \langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}\}, \qquad H := \{\boldsymbol{v} \in \mathbb{Z}_Q^n : \langle \mathbf{b}, \boldsymbol{v} \rangle \equiv 0 \pmod{Q}\}.$$

Conditioned on R, the reduced outcome $\boldsymbol{u} \bmod Q \in \mathbb{Z}_Q^n$ is exactly uniform on H. In particular, $\Pr[\boldsymbol{u} \bmod Q = \mathbf{0} \mid R] = 1/|H| = Q^{-(n-1)}$.

Proof. Fix E. Apply $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the chirp-free state (1.2). For each $\boldsymbol{u} \in \mathbb{Z}_M^n$, the (unnormalized) amplitude equals

$$\omega_M^{\langle \boldsymbol{u}, \boldsymbol{v}^* \rangle} \cdot \sum_{j \in \mathbb{Z}} \sum_{\boldsymbol{k} \in \mathcal{K}} \alpha_E(j, \boldsymbol{k})\, i^{\|\boldsymbol{k}\|^2}\, \omega_M^{j \langle \boldsymbol{u}, \Delta \rangle + M_2 \langle \boldsymbol{u}, \boldsymbol{k} \rangle},$$

so the outcome probabilities are proportional to the squared magnitude of the inner sum (the global phase $\omega_M^{\langle \boldsymbol{u}, \boldsymbol{v}^* \rangle}$ disappears).

Because $M_2 = M/2$, we have $\omega_M^{M_2 \langle \boldsymbol{u}, \boldsymbol{k} \rangle} = (-1)^{\langle \boldsymbol{u}, \boldsymbol{k} \rangle}$, which depends only on the parity pattern of $\boldsymbol{u}_{[2..n]}$ since $k_1 = 0$ for all $\boldsymbol{k} \in \mathcal{K}$. On the event R we have $\langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}$, hence

$$\langle \boldsymbol{u}, \Delta \rangle = 2D^2 \langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{2D^2 Q} = 0 \pmod{M},$$

and therefore $\omega_M^{j\langle \boldsymbol{u},\Delta\rangle} = 1$ for all j. Therefore, conditioned on R, the outcome probabilities depend on $\boldsymbol{u}$ only through the parity pattern of $\boldsymbol{u}_{[2..n]}$.

Now fix any $\boldsymbol{v} \in H$. The fibre $F_{\boldsymbol{v}} := \{\boldsymbol{u} \in \mathbb{Z}_M^n : \boldsymbol{u} \bmod Q = \boldsymbol{v}\}$ consists of the vectors $\boldsymbol{u} = \boldsymbol{v} + Q\boldsymbol{t}$ with $\boldsymbol{t} \in \mathbb{Z}_{M/Q}^n = \mathbb{Z}_{2D^2}^n$. Since Q is odd, as $\boldsymbol{t}$ ranges over $\mathbb{Z}_{2D^2}^n$ the parity pattern of $(\boldsymbol{v} + Q\boldsymbol{t})_{[2..n]}$ is exactly uniform over $\{0,1\}^{n-1}$, independently of $\boldsymbol{v}$. Hence each parity pattern occurs in $F_{\boldsymbol{v}}$ with the same multiplicity for every $\boldsymbol{v} \in H$, and since the conditional probabilities depend only on this parity pattern, $\mu_E[F_{\boldsymbol{v}} \mid R]$ is constant over $\boldsymbol{v} \in H$. This proves uniformity of $\boldsymbol{u} \bmod Q$ on H conditioned on R. Finally, because $b_1 = -1$ is a unit in $\mathbb{Z}_Q$, $|H| = Q^{n-1}$, so $\Pr[\boldsymbol{u} \bmod Q = \boldsymbol{0} \mid R] = Q^{-(n-1)}$. □

Proposition 2.3 (Canonical-gauge compilation given coherent transcript access)**.** Assume AC1 and suppose the state-preparation procedure is given as an explicit circuit that admits a deferred-measurement implementation producing a coherent transcript register E from which $v_1^*(E) \bmod M_2$ (equivalently, the gauge parameter $\lambda(E)$ defined in AC4) can be computed reversibly. Then one can compile the preparation by appending an E-controlled basis translation on X_1 by a multiple of D^2 (modulo M) so that, conditioned on E, the resulting offset satisfies the canonical-gauge requirement $v_1^*(E) \equiv \mathrm{ctr}(X_1 \bmod D^2) \pmod{M_2}$ (equivalently, $\lambda(E) = 0$). If the preparation is a black-box state oracle that does not expose such coherent access to $\lambda(E)$, the canonical-gauge condition is an additional oracle promise and is not implied by AC1 alone.

Proof. Fix a transcript E. By AC1 we have $X_1 \equiv -2D^2 j + v_1^*(E) \pmod{M_2}$ with $M_2 = D^2 Q$. Let $\tau(E) := v_1^*(E) \bmod D^2$ and let $\bar{v}_1 := \mathrm{ctr}(\tau(E)) \in (-D^2/2, D^2/2]$. Because $D^2 \mid M_2$, there exists a unique $\lambda(E) \in \mathbb{Z}_Q$ such that $v_1^*(E) \equiv \bar{v}_1 + \lambda(E)D^2 \pmod{M_2}$. By assumption of this proposition, $\lambda(E)$ is a deterministic function of E that can be computed reversibly inside the known preparation circuit. Inside the preparation unitary (where E is available coherently), apply the controlled basis translation $|x\rangle_{X_1} \mapsto |x - \lambda(E)D^2 \bmod M\rangle_{X_1}$. This map is a permutation of the computational basis of X_1 and hence unitary. It leaves $\tau = X_1 \bmod D^2$ invariant, and it replaces the offset representative by $\bar{v}_1$. This translation can therefore be hardwired into the state-preparation circuit to enforce $\lambda(E) = 0$ with only polynomial overhead. □

Lemma 2.4 (Center-referenced chirp cancellation)**.** Assume AC1-AC4. Then the procedure in AC4 defines, for each measured τ, a unitary $U_{\mathrm{corr}}^{(\tau)}$ on the first coordinate register such that $|\varphi_7\rangle := U_{\mathrm{corr}}|\widetilde{\varphi}_7\rangle$ has the form (1.2). The quadratic chirp $e^{-2\pi i j^2/Q}$ in the loop index is removed on every branch.

Proof. Under AC1, we have $X_1 \equiv -2D^2 j + v_1^* \pmod{M_2}$ with v_1^* independent of j. Conditioned on the transcript E, the measurement of $\tau := X_1 \bmod D^2$ is deterministic and yields $\tau \equiv v_1^* \bmod D^2$ on every branch. By the gauge choice in AC4, with $\bar{v}_1 = \mathrm{ctr}(\tau)$ and $v_{1,Q} = \bar{v}_1 \bmod Q$, we have $(X_1 \bmod Q) - v_{1,Q} \equiv -2D^2 j \pmod{Q}$ on every branch. The reversible computation of j_x in AC4 is well-defined modulo Q by AC2-AC3. For $x = X_1$ on a branch with loop label j we have $j_x \equiv j \pmod{Q}$. The unitary U_{corr} multiplies that branch by $\exp(2\pi i j^2/Q)$. It cancels the original chirp $e^{-2\pi i j^2/Q}$. The map is diagonal in the computational basis of X_1. Hence it is unitary. □

Remark 2.5 (Justification of Additional Condition AC5)**.** Additional Condition AC5 isolates the only additional front-end property used by Step $9^\dagger$: spectral concentration of the one-dimensional Karst-wave envelope at the resonant frequencies t/Q with $t \equiv 0 \pmod{Q}$.

In Chen's Karst-wave front end, conditioned on a transcript E, the post-Step 7 amplitudes factor as $\alpha_E(j, \boldsymbol{k}) = \alpha_E(j)\beta_E(\boldsymbol{k})$ (see Eq. (35) in Chen [2024]). After chirp cancellation (Additional

Condition AC4), the Fourier amplitude at $\boldsymbol{u} \in \mathbb{Z}_M^n$ factorizes as

$$S_E(\boldsymbol{u}) = \widehat{\alpha}_E(t(\boldsymbol{u})/Q) \cdot G_E(\boldsymbol{u}),$$

where $G_E(\boldsymbol{u})$ collects the grid contribution. Here we write

$$t(\boldsymbol{u}) \;:=\; \langle \mathbf{b}, \boldsymbol{u} \rangle \bmod Q \;\in\; \mathbb{Z}_Q,$$

and identify $t(\boldsymbol{u})$ with its standard representative in $\{0, \ldots, Q-1\}$ when used as an argument of $\widehat{\alpha}_E(\cdot)$ on $[0, 1)$. Taking the envelope width $\sigma_J \gtrsim Q \log n$ yields $|\widehat{\alpha}_E(t/Q)| \leq \exp(-\Omega(\log^2 n)) \cdot |\widehat{\alpha}_E(0)|$ for all integers $t \not\equiv 0 \pmod{Q}$ by standard Poisson-summation estimates. Since $\sum_{t \in \mathbb{Z}_Q} |\widehat{\alpha}_E(t/Q)|^2 \geq |\widehat{\alpha}_E(0)|^2$, this implies the leakage bound (2.1).

Remark on linear phases. A time shift $j \mapsto j - j_0$ multiplies $\widehat{\alpha}_E(\theta)$ by a unit-modulus factor and does not affect its magnitude. In contrast, a nontrivial linear phase $\alpha_E(j) \mapsto \alpha_E(j) e^{2\pi i \nu j}$ shifts the spectrum $\widehat{\alpha}_E(\theta) \mapsto \widehat{\alpha}_E(\theta + \nu)$ and can move the dominant residue class away from $t = 0$ on the grid $\{t/Q\}$. In our setting, such a run-dependent shift is exactly the residual $\omega_Q^{-\lambda(E)j}$ phase arising from a gauge mismatch in AC4; the canonical-gauge promise rules it out.

No additional near-uniformity assumption is needed for the *reduced* samples: Lemma 2.2 shows that conditioned on $\langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}$, the reduction $\boldsymbol{u} \bmod Q$ is exactly uniform on the dual hyperplane H, and $\Pr[\boldsymbol{u} \bmod Q = \mathbf{0} \mid \langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0] = 1/|H| = Q^{-(n-1)}$.

Proposition 2.6 (Spectral concentration for the re-tuned Karst wave)**.** For the complex-Gaussian Karst-wave front end of Chen [2024] (where, conditioned on a transcript E, the post-Step 7 amplitudes factor as $\alpha_E(j, \boldsymbol{k}) = \alpha_E(j)\beta_E(\boldsymbol{k})$), instantiated with any re-tuned configuration that satisfies Definition 2.1, Additional Condition AC5 holds (i.e., the leakage bound (2.1) fails only with probability $\mathrm{negl}(n)$ over the front-end transcript).

Proof. Under Chen's Karst-wave analysis, conditioned on a transcript E the loop envelope $\alpha_E(j)$ is (up to a global phase, an integer shift in j, and negligible truncation) a discrete Gaussian of width σ_J, and the factorization $\alpha_E(j, \boldsymbol{k}) = \alpha_E(j)\beta_E(\boldsymbol{k})$ holds. An integer shift in j does not affect $|\widehat{\alpha}_E(\cdot)|$. Any additional run-dependent linear phase on the $1/Q$ grid would shift the dominant residue class away from $t = 0$; in our setting this is precisely the gauge-mismatch phenomenon discussed and ruled out in AC4. Thus it suffices to bound the centered discrete-Gaussian prototype $g_\sigma(j) := \exp(-\pi j^2/\sigma^2)$. Its Fourier series satisfies the Poisson-summation identity

$$\sum_{j \in \mathbb{Z}} g_\sigma(j)\, e^{2\pi i j \theta} = \sigma \sum_{m \in \mathbb{Z}} \exp\bigl(-\pi \sigma^2 (\theta - m)^2\bigr).$$

Evaluating at $\theta = t/Q$, for any integer $t \not\equiv 0 \pmod{Q}$ let $m^\star \in \mathbb{Z}$ be a nearest integer to t/Q (so $m^\star \in \{0, 1\}$). Then $|t/Q - m^\star| \geq 1/Q$, hence the dominant Gaussian term in the Poisson sum is at most $\sigma \exp(-\pi\sigma^2/Q^2)$, and all other terms are smaller by an additional factor $\exp(-\Omega(\sigma^2))$. Consequently, for some absolute constant $c_0 > 0$,

$$|\widehat{\alpha}_E(t/Q)| \leq |\widehat{\alpha}_E(0)| \cdot \exp\Bigl(-c_0 \cdot (\sigma_J/Q)^2\Bigr) \quad \text{for all } t \not\equiv 0 \pmod{Q},$$

up to negligible truncation error from the effective window. With $\sigma_J \geq C_{\mathrm{spec}} Q \log n$ (Definition 2.1), this gives $|\widehat{\alpha}_E(t/Q)| \leq |\widehat{\alpha}_E(0)| \exp(-\Omega(\log^2 n))$. Since $Q = \mathrm{poly}(n)$ in the re-tuned regime, summing the squared bound over the $(Q-1)$ nonzero residues yields

$$\sum_{t \in \mathbb{Z}_Q \setminus \{0\}} |\widehat{\alpha}_E(t/Q)|^2 \leq (Q-1) \exp(-\Omega(\log^2 n)) \cdot |\widehat{\alpha}_E(0)|^2.$$

Using $\sum_{t\in\mathbb{Z}_Q} |\widehat{\alpha}_E(t/Q)|^2 \geq |\widehat{\alpha}_E(0)|^2$ proves (2.1). $\square$

2.2 Access model

Definition 2.7 (Access model)**.** We assume (circuit-level) quantum access to an outcome-conditioned state-preparation procedure that, on each invocation, outputs one copy of the post-Step 7 state $|\widetilde{\varphi}_7\rangle$ on coordinate registers $X \in (\mathbb{Z}_M)^n$. We additionally assume a *canonical-gauge* implementation as in Additional Condition AC4: inside the preparation circuit the first coordinate is translated by an E-dependent multiple of D^2 (modulo M) so that, conditioned on E, writing $\tau := X_1 \bmod D^2$ and $\bar{v}_1 := \mathrm{ctr}(\tau)$, the offset satisfies $v_1^*(E) \equiv \bar{v}_1 \pmod{M_2}$ (equivalently, $\lambda(E) = 0$ in Additional Condition AC4). Consequently, measuring τ allows the post-processing to compute $v_{1,Q} = \bar{v}_1 \bmod Q$ and the chirp-correction phase for that run.

For each invocation there exists an unobserved classical transcript E such that, conditioned on E, the reduced state on X satisfies Additional Condition AC1 with offset $\boldsymbol{v}^*(E)$ and amplitudes $\alpha_E(j, \boldsymbol{k})$. Across invocations the transcript is re-sampled, so the offset $\boldsymbol{v}^*(E)$ may change from run to run and cannot be forced to repeat.

The post-processing is allowed to apply any quantum circuit of size poly(n) to the output registers, including arithmetic in $\mathbb{Z}_M$ and $\mathbb{Z}_Q$, intermediate measurements, and $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$. We do *not* assume the ability to query the preparation on chosen basis inputs j, to classically compute the full offset $\boldsymbol{v}^*(E)$, or to obtain two copies of $|\widetilde{\varphi}_7\rangle$ with the same offset.

In this model, an algorithm acts on the superposition directly. The offset $\boldsymbol{v}^*(E)$ depends on the run-specific outcomes E and behaves like a one-time pad on the coordinate values in the computational basis. Step $9^\dagger$ shows that this offset turns into a harmless phase pattern after a Fourier transform. We can then sample from the dual lattice without ever learning Δ or $\boldsymbol{v}^*$ as explicit classical data.

Lemma 2.8 (Offset coherence)**.** In the deferred-measurement unitary of Chen [2024], take the program point just before Step 8 and after modulus splitting and center extraction (Steps 5-7). For each fixed transcript E, the post-Step 7 coordinate block in $(\mathbb{Z}_M)^n$ is supported on

$$\left\{\, j\Delta + \boldsymbol{v}^*(E) + M_2\boldsymbol{k} \bmod M : j \in \mathbb{Z},\ \boldsymbol{k} \in \mathcal{K} \,\right\},$$

for some index set $\mathcal{K} \subseteq \{0\} \times \mathbb{Z}^{n-1}$, where $\Delta = 2D^2\mathbf{b}$ and the offset $\boldsymbol{v}^*(E) \in (\mathbb{Z}_M)^n$ depends on E but not on j. Reducing modulo M_2 eliminates the $M_2\boldsymbol{k}$ term and yields

$$\mathbf{X}(j) \equiv 2D^2 j\,\mathbf{b} + \boldsymbol{v}^*(E) \pmod{M_2}.$$

Proof. This is immediate from Chen's post-Step 7 computational-basis form (see the excerpted Eq. (14)): the basis values are $2Dj\,\boldsymbol{x} + \boldsymbol{v}' + \frac{M}{2}\boldsymbol{k} \bmod M$ with $\boldsymbol{k} \in 0\,|\,\mathbb{Z}^{n-1}$ and $\boldsymbol{x} = D\mathbf{b}$. Substituting $\boldsymbol{x} = D\mathbf{b}$ gives $2Dj\,\boldsymbol{x} = 2D^2 j\,\mathbf{b} = j\Delta$ with $\Delta = 2D^2\mathbf{b}$, and writing $\boldsymbol{v}^*(E) := \boldsymbol{v}' \bmod M$ yields the claimed support in $(\mathbb{Z}_M)^n$. Reducing modulo $M_2 = M/2$ eliminates the $M_2\boldsymbol{k}$ term, giving $\mathbf{X}(j) \equiv 2D^2 j\,\mathbf{b} + \boldsymbol{v}^*(E) \pmod{M_2}$. $\square$

3 The new Step $9^\dagger$

Algorithm 1 Step $9^\dagger$

Input: A lattice instance $L_q^\perp(\mathbf{A})$ arising from a chosen-secret LWE instance as in Chen [2024] (i.e., $\mathbf{A} = [2p_1\mathbf{t} \mid \mathbf{U}^\mathsf{T} \mid \mathbf{I}_m]$) together with public parameters $(q, \beta, p_1, \ldots, p_\kappa)$.
Oracle: State-preparation access to $|\widetilde{\varphi}_7\rangle$ as in Definition 2.7.
Parameters: Re-tuned Karst-wave configuration $(c, D, r, s, t, u, P, M, Q)$ (see Remark 3.1).
Output: Either the LWE secret s and error e, or FAIL (meaning the current norm guess is rejected).
Fix absolute constants $C_{\text{rank}} > 1$, $C_{\text{try}} \geq 1$, and $C_{\text{samp}} \geq 1$.
Set the target sample count $N \leftarrow \lceil C_{\text{rank}}(n-1) \rceil$ and the batch budget $T_{\max} \leftarrow \lceil C_{\text{try}} n \rceil$.
Set a per-batch oracle-call cap $B_{\max} \leftarrow \lceil C_{\text{samp}} \cdot N \rceil$.
Precompute inv $:= (-2D^2)^{-1} \in \mathbb{Z}_Q$ ▷ Exists by AC2-AC3. Used inside AC4.
for trial $= 1$ **to** $T_{\max}$ **do**
    Initialize $\mathcal{S} \leftarrow \emptyset$ and calls $\leftarrow 0$.
    while $|\mathcal{S}| < N$ **and** calls $< B_{\max}$ **do**
        calls $\leftarrow$ calls $+ 1$.
        $\boldsymbol{u}_Q \leftarrow$ Step9DaggerSample(D, M, Q, inv) ▷ Algorithm 2.
        if $\boldsymbol{u}_Q \neq \mathbf{0}$ **then**
            Append $\boldsymbol{u}_Q$ as a row of $\mathcal{S}$.
        end if
    end while
    if $|\mathcal{S}| < N$ **then**
        continue ▷ Sampling cap exceeded (wrong guess or atypical transcript).
    end if
    $\mathbf{b}_Q \leftarrow$ RecoverBModQ$(\mathcal{S}, Q)$ ▷ Algorithm 4.
    if $\mathbf{b}_Q =$ FAIL **then**
        continue
    end if
    $(s, e) \leftarrow$ ExtractAndVerifyLWE$(\mathbf{b}_Q, \mathbf{A}, q, \beta, p_1, \ldots, p_\kappa)$ ▷ Algorithm 3.
    if $(s, e) \neq$ FAIL **then**
        return (s, e)
    end if
end for
return FAIL

Algorithm 2 STEP9DAGGERSAMPLE: chirp correction and direct Fourier sampling

Input: Parameters (D, M, Q) and $\mathrm{inv} := (-2D^2)^{-1} \in \mathbb{Z}_Q$.
Oracle: One copy of $|\widetilde{\varphi}_7\rangle$ on coordinate registers $X \in (\mathbb{Z}_M)^n$ (Definition 2.7).
Output: A reduced sample $\boldsymbol{u}_Q \in \mathbb{Z}_Q^n$ (possibly $\mathbf{0}$).
Query the state-preparation oracle to obtain $|\widetilde{\varphi}_7\rangle$ on X.
Measure $\tau \leftarrow X_1 \bmod D^2$ ▷ Deterministic conditioned on the transcript by AC1/AC4.
Compute $v_{1,Q} \leftarrow \mathrm{ctr}(\tau) \bmod Q$.
Apply the τ-controlled diagonal unitary $U_{\mathrm{corr}}^{(\tau)}$ on X_1 as defined in AC4 (using inv and $v_{1,Q}$).
Apply $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to X and measure $\boldsymbol{u} \in \mathbb{Z}_M^n$.
return $\boldsymbol{u}_Q \leftarrow \boldsymbol{u} \bmod Q$.

Algorithm 3 EXTRACTANDVERIFYLWE: parse and certify (s, e)

Input: $\mathbf{b}_Q \in \mathbb{Z}_Q^n$, the defining matrix $\mathbf{A}$ of $L_q^{\perp}(\mathbf{A})$, and public parameters $(q, \beta, p_1, \ldots, p_\kappa)$.
Output: The LWE secret s and error e, or FAIL.
Let $m \leftarrow$ the number of rows of $\mathbf{A}$ and $n \leftarrow$ the number of columns of $\mathbf{A}$.
Let $\ell \leftarrow n - m - 1$ (so that $\mathbf{A}$ has block form $[2p_1\mathbf{t} \mid \mathbf{U}^\top \mid \mathbf{I}_m]$ with $\mathbf{U} \in \mathbb{Z}_q^{\ell \times m}$ in the chosen-secret embedding).
Lift $\mathbf{b}_Q$ coordinate-wise to $\mathbf{b} \in (-Q/2, Q/2]^n \cap \mathbb{Z}^n$.
if $b_1 \neq -1$ as an integer **then**
  return FAIL
end if
for $i = 2$ **to** n **do**
  if $2p_1 \nmid b_i$ in $\mathbb{Z}$ **then**
    return FAIL
  end if
end for
for $\eta = 2$ **to** κ **do**
  if $b_\eta \neq 2p_1p_\eta$ in $\mathbb{Z}$ **then**
    return FAIL
  end if
end for
Parse $\mathbf{b} = [-1,\ 2p_1s^\top,\ 2p_1e^\top]^\top$ with $s \in \mathbb{Z}^\ell$ and $e \in \mathbb{Z}^m$, and recover s, e by integer division by $2p_1$.
Check shortness (e.g. $\|s\|_\infty, \|e\|_\infty \leq \beta \log n$) and any promised norm bounds for $\|\mathbf{b}\|$ from the lattice analysis.
if $\mathbf{Ab} \not\equiv \mathbf{0} \pmod q$ **then**
  return FAIL
end if
return (s, e).

Algorithm 4 RecoverBModQ: solve for $\mathbf{b} \bmod Q$ from dual samples

Input: A matrix $\mathcal{S} \in (\mathbb{Z}_Q)^{N\times n}$ whose rows are nonzero samples $\boldsymbol{u}_Q$.
Output: $\mathbf{b}_Q = \mathbf{b} \bmod Q$, or FAIL.
Let $U \in (\mathbb{Z}_Q)^{N\times(n-1)}$ be columns $2..n$ of $\mathcal{S}$, and let $\boldsymbol{y} \in (\mathbb{Z}_Q)^N$ be column 1 of $\mathcal{S}$.
Solve $U\boldsymbol{z} \equiv \boldsymbol{y} \pmod Q$ and test uniqueness (e.g. via Smith normal form).
if no solution exists **or** the solution is not unique **then**
    return FAIL
end if
Let $\boldsymbol{z}$ be the unique solution and set $\mathbf{b}_Q \leftarrow [-1, \boldsymbol{z}^\top]^\top \in \mathbb{Z}_Q^n$.
return $\mathbf{b}_Q$.

Remark 3.1 (Classical guessing loop for the norm)**.** As in Chen [2024], the small modulus is chosen as $Q = (c{+}1)\|\mathbf{b}\|^2$, where $\mathbf{b}$ is the (unknown) unique shortest vector of $L_q^\perp(\mathbf{A})$. We run a classical outer loop over all polynomially many candidates B_{guess} for $\|\mathbf{b}\|^2$. For each guess we set $Q \leftarrow (c{+}1)B_{\text{guess}}$, choose an odd D with $\gcd(D, Q) = 1$, derive $M = 2D^2Q$ (hence $M_2 = D^2Q$), and complete the remaining Karst-wave parameters as in Definition 2.1. We then execute Algorithm 1 for this guess.

Only the correct guess yields dual samples supported on the hyperplane $\langle \mathbf{b}, \boldsymbol{u}\rangle \equiv 0 \pmod Q$ and passes the explicit verification in Algorithm 3, except with negligible probability. To keep rejection for incorrect guesses polynomial-time, Algorithm 1 uses a fixed trial budget $T_{\max} = \text{poly}(n)$ and a per-trial sampling cap $B_{\max} = \text{poly}(n)$, returning FAIL if no verified solution is found within budget.

Remark 3.2 (Instantiation for LWE)**.** For the q-ary lattice $L_q^\perp(\mathbf{A})$ in Chen [2024], the promised unique shortest vector has the form

$$\mathbf{b} = [-1,\, 2p_1 s^\top,\, 2p_1 e^\top]^\top \in \mathbb{Z}^n,$$

where s is the LWE secret and e is the error vector. Because $\|\mathbf{b}\|_\infty \le 2p_1\beta \log n < Q/2$ in the parameter regime of interest, the reduction map $\mathbf{b} \mapsto \mathbf{b} \bmod Q$ is injective on the centred cube that contains the true shortest vector. Therefore, once Algorithm 1 recovers $\mathbf{b} \bmod Q$, the canonical lift to $(-Q/2, Q/2]^n$ recovers $\mathbf{b}$ as an integer vector. Dividing the last $n-1$ coordinates by $2p_1$ (after checking divisibility) yields s and e, and the check $\mathbf{A}\mathbf{b} \equiv 0 \pmod q$ certifies correctness.

Fourier sampling. We apply $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the coordinate register block and then measure $\boldsymbol{u} \in \mathbb{Z}_M^n$ (Algorithm 2). The outcome distribution is analyzed in the next part.

3.1 Correctness under Additional Conditions

Lemma 3.3 (Dual sampling via direct QFT)**.** Assume AC1-AC5 and let $|\varphi_7\rangle$ be the chirp-free state from Lemma 2.4. Fix a good measurement transcript E (in the sense of Additional Condition AC5) and write the conditional state as

$$|\varphi_7\rangle = \sum_{j\in\mathbb{Z}} \sum_{\boldsymbol{k}\in\mathcal{K}} \alpha_E(j, \boldsymbol{k})\, i^{\|\boldsymbol{k}\|^2}\ |j\Delta + \boldsymbol{v}^* + M_2\boldsymbol{k} \bmod M\rangle\,,$$

where $\Delta = 2D^2\mathbf{b}$ and $M_2 = M/2$. After applying $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the coordinate registers and measuring $\boldsymbol{u} \in \mathbb{Z}_M^n$, we have

$$\Pr\big[\langle \mathbf{b}, \boldsymbol{u}\rangle \equiv 0 \pmod Q\big] \ge 1 - \text{negl}(n).$$

Moreover, by Lemma 2.2, conditioned on the event $\langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}$ the reduced outcome $\boldsymbol{u} \bmod Q$ is exactly uniform on the dual hyperplane H, hence $\Pr[\boldsymbol{u} \bmod Q = \mathbf{0}] \le Q^{-(n-1)} + \mathrm{negl}(n)$.

Proof of Lemma 3.3. Fix E. Apply $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ to the chirp-free state (1.2). For each $\boldsymbol{u} \in \mathbb{Z}_M^n$, the (unnormalized) amplitude equals

$$\omega_M^{\langle \boldsymbol{u}, \boldsymbol{v}^* \rangle} \cdot \sum_{j \in \mathbb{Z}} \sum_{\boldsymbol{k} \in \mathcal{K}} \alpha_E(j, \boldsymbol{k})\, i^{\|\boldsymbol{k}\|^2}\, \omega_M^{j\langle \boldsymbol{u}, \Delta \rangle + M_2 \langle \boldsymbol{u}, \boldsymbol{k} \rangle},$$

so the outcome probabilities are proportional to the squared magnitude of the inner sum (the global phase $\omega_M^{\langle \boldsymbol{u}, \boldsymbol{v}^* \rangle}$ disappears).

By AC5, conditioned on E the coefficients factor as $\alpha_E(j, \boldsymbol{k}) = \alpha_E(j)\beta_E(\boldsymbol{k})$. Using $M_2 = M/2$ we have $\omega_M^{M_2 \langle \boldsymbol{u}, \boldsymbol{k} \rangle} = (-1)^{\langle \boldsymbol{u}, \boldsymbol{k} \rangle}$, and since $k_1 = 0$ this depends only on the parity pattern of $\boldsymbol{u}_{[2..n]}$. Moreover $M = 2D^2 Q$ and $\Delta = 2D^2 \mathbf{b}$, hence

$$\omega_M^{j\langle \boldsymbol{u}, \Delta \rangle} = \exp\left(2\pi i\, j\, \frac{t(\boldsymbol{u})}{Q}\right),$$

where $t(\boldsymbol{u}) := \langle \mathbf{b}, \boldsymbol{u} \rangle \bmod Q \in \mathbb{Z}_Q$ and we identify $t(\boldsymbol{u})$ with its representative in $\{0, \ldots, Q-1\}$ when used inside the exponential. Therefore $S_E(\boldsymbol{u})$ factorizes as

$$S_E(\boldsymbol{u}) = \Big(\sum_{j \in \mathbb{Z}} \alpha_E(j)\, e^{2\pi i j\, t(\boldsymbol{u})/Q}\Big) \cdot \Big(\sum_{\boldsymbol{k} \in \mathcal{K}} \beta_E(\boldsymbol{k})\, i^{\|\boldsymbol{k}\|^2}\, (-1)^{\langle \boldsymbol{u}, \boldsymbol{k} \rangle}\Big) = \widehat{\alpha}_E(\langle \mathbf{b}, \boldsymbol{u} \rangle / Q) \cdot G_E(\boldsymbol{u}),$$

where $G_E(\boldsymbol{u})$ depends only on the parity pattern of $\boldsymbol{u}_{[2..n]}$. Thus

$$\Pr[\boldsymbol{u}] \;\propto\; |S_E(\boldsymbol{u})|^2 = |\widehat{\alpha}_E(\langle \mathbf{b}, \boldsymbol{u} \rangle / Q)|^2 \cdot |G_E(\boldsymbol{u})|^2.$$

For each residue $t \in \mathbb{Z}_Q$, let

$$T_t := \{\boldsymbol{u} \in \mathbb{Z}_M^n : \langle \mathbf{b}, \boldsymbol{u} \rangle \equiv t \pmod{Q}\}.$$

Because $b_1 = -1$ is a unit in $\mathbb{Z}_Q$, for every fixed $\boldsymbol{u}_{[2..n]} \in \mathbb{Z}_M^{n-1}$ and every t there are exactly $M/Q = 2D^2$ values of $u_1 \in \mathbb{Z}_M$ such that $\boldsymbol{u} \in T_t$. Since $G_E(\boldsymbol{u})$ does not depend on u_1, the total grid mass $\sum_{\boldsymbol{u} \in T_t} |G_E(\boldsymbol{u})|^2$ is the same for every t. Hence the total probability mass of T_t is proportional to $|\widehat{\alpha}_E(t/Q)|^2$, up to negligible truncation error.

By the leakage bound (2.1) (good transcript) and this proportionality, the total off-resonant probability mass satisfies

$$\sum_{t \neq 0} \Pr[T_t] = \frac{\sum_{t \neq 0} |\widehat{\alpha}_E(t/Q)|^2}{\sum_{t \in \mathbb{Z}_Q} |\widehat{\alpha}_E(t/Q)|^2} \le \mathrm{negl}(n),$$

so

$$\Pr\big[\langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}\big] \ge 1 - \mathrm{negl}(n).$$

Finally, conditioned on this resonance event, Lemma 2.2 gives exact uniformity of $\boldsymbol{u} \bmod Q$ on H, and in particular $\Pr[\boldsymbol{u} \bmod Q = \mathbf{0}] \le Q^{-(n-1)} + \mathrm{negl}(n)$. □

Theorem 3.4 (Step 9† is correct)**.** Assume AC1-AC5 and run Algorithm 1 in the correct iteration of the outer norm-guessing loop. Each sample $\boldsymbol{u}_Q \in \mathbb{Z}_Q^n$ appended to $\mathcal{S}$ by Algorithm 1 is non-zero and satisfies

$$\langle \mathbf{b}, \boldsymbol{u}_Q \rangle \equiv 0 \pmod{Q}$$

except with probability $\mathrm{negl}(n)$ over the choice of a good transcript and the internal randomness of the measurements. Solving the resulting linear system modulo Q (using the known normalization $b_1 = -1$) recovers $\mathbf{b} \bmod Q$ and hence $\mathbf{b}$ as an integer vector. The LWE secret and error are then extracted and verified in polynomial time with overwhelming probability.

Proof. By Additional Condition AC5, a transcript drawn from the front end is good except with probability $\mathrm{negl}(n)$. Algorithm 1 makes only $\mathrm{poly}(n)$ oracle calls, so by a union bound we may condition on the event that all invocations are good, losing at most a negligible term in the overall success probability.

In each oracle call within Algorithm 1, the sampling primitive Step9DaggerSample (Algorithm 2) measures $\tau = X_1 \bmod D^2$ and applies the diagonal gate $U_{\mathrm{corr}}^{(\tau)}$. Lemma 2.4 (which uses the canonical-gauge requirement in AC4) shows that this removes the quadratic chirp and produces the chirp-free state $|\varphi_7\rangle$ assumed in AC5.

Applying $\mathrm{QFT}_{\mathbb{Z}_M}^{\otimes n}$ and measuring yields $\boldsymbol{u} \in \mathbb{Z}_M^n$. Lemma 3.3 implies that $\langle \mathbf{b}, \boldsymbol{u} \rangle \equiv 0 \pmod{Q}$ except with probability $\mathrm{negl}(n)$, and Lemma 2.2 implies that, conditioned on this resonance event, $\boldsymbol{u} \bmod Q$ is exactly uniform on $H = \{\boldsymbol{v} \in \mathbb{Z}_Q^n : \langle \mathbf{b}, \boldsymbol{v} \rangle \equiv 0 \pmod{Q}\}$. In particular, $\Pr[\boldsymbol{u} \bmod Q = \mathbf{0}] \le Q^{-(n-1)} + \mathrm{negl}(n)$, so the rejection step $\boldsymbol{u}_Q \neq \mathbf{0}$ fails only negligibly often. A union bound over the $N = O(n)$ accepted samples in a batch implies that, with probability $1 - \mathrm{negl}(n)$, every accepted $\boldsymbol{u}_Q$ satisfies $\langle \mathbf{b}, \boldsymbol{u}_Q \rangle \equiv 0 \pmod{Q}$.

Fix such a successful batch. Because $b_1 = -1$ is a unit in $\mathbb{Z}_Q$, the projection $H \to \mathbb{Z}_Q^{n-1}$ given by $\boldsymbol{u} \mapsto \boldsymbol{u}_{[2..n]}$ is a bijection. Hence the accepted vectors $\boldsymbol{u}_{Q,[2..n]}$ are i.i.d. (up to a negligible statistical error from the rare off-resonant outputs) uniform over $\mathbb{Z}_Q^{n-1} \setminus \{\mathbf{0}\}$. Let $U \in \mathbb{Z}_Q^{N \times (n-1)}$ be the matrix with these rows and let $\boldsymbol{y}$ be the vector of first coordinates. Then the true unknown $\mathbf{b}_{[2..n]} \bmod Q$ satisfies $U\mathbf{b}_{[2..n]} \equiv \boldsymbol{y} \pmod{Q}$.

With overwhelming probability U is injective as a map $(\mathbb{Z}_Q)^{n-1} \to (\mathbb{Z}_Q)^N$. Indeed, for any prime $p \mid Q$, the reduction $U \bmod p$ is (up to a negligible error from excluding the all-zero row) an i.i.d. uniform random matrix over the field $\mathbb{Z}_p$ with $N \ge C_{\mathrm{rank}}(n-1)$ rows, and standard rank bounds give $\Pr[\mathrm{rank}_{\mathbb{Z}_p}(U \bmod p) < n-1] \le p^{-\Omega(n)}$. If $\mathrm{rank}_{\mathbb{Z}_p}(U \bmod p) = n-1$ and $U\boldsymbol{z} \equiv \mathbf{0} \pmod{p^e}$, then reducing modulo p gives $\boldsymbol{z} \equiv \mathbf{0} \pmod{p}$; writing $\boldsymbol{z} = p\boldsymbol{z}_1$ and iterating shows $\boldsymbol{z} \equiv \mathbf{0} \pmod{p^e}$, so $\ker(U \bmod p^e) = \{\mathbf{0}\}$. By the Chinese remainder theorem, $\ker(U \bmod Q) = \{\mathbf{0}\}$. Therefore the system has a unique solution $\boldsymbol{z} \in \mathbb{Z}_Q^{n-1}$, which Algorithm 4 computes (e.g. via Smith normal form) and uses to set $\mathbf{b}_Q = [-1, \boldsymbol{z}^\top]^\top = \mathbf{b} \bmod Q$.

Finally, the bound $\|\mathbf{b}\|_\infty < Q/2$ from (R1) makes the centered lift from $\mathbb{Z}_Q^n$ to $(-Q/2, Q/2]^n$ injective on the relevant region, so lifting $\mathbf{b}_Q$ recovers $\mathbf{b}$ as an integer vector. The subsequent divisibility, planted-prefix, shortness, and lattice-membership checks (Algorithm 3) succeed for the correct guess and reject incorrect guesses, as in Chen [2024]. □

Corollary 3.5 (Conditional quantum algorithm for LWE)**.** Assume the LWE-to-lattice reduction and the unique-shortest-vector promise for $L_q^\perp(\mathbf{A})$ from Chen [2024]. If the corresponding front end can be compiled in a re-tuned configuration satisfying Additional Conditions AC1-AC5, then

Algorithm 1 solves the chosen-secret LWE instance encoded by $L_q^{\perp}(\mathbf{A})$ in quantum polynomial time with overwhelming probability. Via the classical reductions of Chen [2024], this yields a quantum algorithm for standard LWE in the same parameter regime, conditional on AC1-AC5.

4 Conclusion

In this work, we present a new Step $9^{\dagger}$, which turns unknown offsets into harmless phase patterns. Under explicit front-end conditions on the Karst wave window, it yields modular linear relations from direct Fourier sampling on the coordinate registers. We instantiate this step with the parameters and the lattice $L_q^{\perp}(\mathbf{A})$ from Chen [2024]. In the access model of Section 2.2, such a construction gives a direct path from a superposition with support $\{j\Delta + \boldsymbol{v}^*\}$ to samples from the dual lattice. Under these assumptions, we obtain a quantum algorithm that solves the chosen-secret LWE problem of Chen [2024] in almost the same parameter regime as in that work.

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References

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